

Prit Kumar

Systems & AI Security Developer · Open-Source Builder

Bihar, India

✉ pritform@gmail.com ✉ pritr ✉ Prit Kumar

Summary

Systems and open-source developer (B.Tech CSE) building privacy-respecting platforms, local-first distributed architectures, and AI security runtime tooling. Maintainer of **Predirect** (270+ ★ on GitHub) and creator of **sentinel-mcp** (Go capability proxy for Model Context Protocol) and **dhvani-eval** (vernacular voice AI eval harness). Founding tech lead at **Bodhya** FOSS community (aligned with FOSS United & Samagata) mentoring engineering students across Bihar.

Featured Engineering Projects

sentinel-mcp — Model Context Protocol Security Proxy github.com/pritr/sentinel-mcp

Creator & Lead Developer (Go · Systems & AI Security) 2026

- Engineered a zero-overhead (sub-0.2ms latency) capability-gated security reverse proxy in Go for Model Context Protocol (MCP) JSON-RPC execution, sandboxing LLM agent actions.
- Implemented real-time taint analysis and Shannon entropy anomaly detection on tool responses to intercept credential leakage (AWS tokens, private keys) before context ingestion.
- Designed declarative capability policies enforcing directory path whitelisting and SSRF egress blocking (blocking RFC 1918 private subnets and 169.254.169.254 cloud metadata).

dhvani-eval — Indic Voice AI & Telephony Benchmark Suite github.com/pritr/dhvani-eval

Creator & Researcher (Python · Speech AI & Audio DSP) 2026

- Architected an open-source evaluation suite for Indic voice models, benchmarking Devanagari Unicode normalization, character/word error rates (CER/WER), and code-mixed Hinglish alignment.
- Built an acoustic telephony degradation engine simulating rural Indian network conditions (8kHz ITU-T bandpass downsampling, burst packet jitter via Gilbert-Elliott modeling, and ambient noise injection).
- Measured streaming voice turnaround metrics (Time to First Token [TTFT], End of Utterance [EOU]) to quantify latency and hallucination degradation under 2G bandwidth.

pebble-sync — Offline-First CRDT Sync Engine github.com/pritr/pebble-sync

Creator (TypeScript · Distributed Systems) 2026

- Developed a zero-dependency state-based CRDT engine supporting hierarchical documents, OR-Sets, and LWW-Registers with mathematically proven eventual consistency convergence.
- Implemented Merkle clock differential replication, slashing wire transfer payloads by >80% over high-latency connections compared to full-state replication.
- Integrated tamper-evident SHA-256 cryptographic hash chains and signature verification for zero-trust edge/mesh synchronization.

Predirect — Privacy Browser Extension github.com/pritr/predirect

Maintainer & Lead Developer (270+ ★ on GitHub · GPL-3.0) Nov 2023 - Present

- Manifest V3 browser extension redirecting 30 tracked sites (YouTube, X, Reddit, Google, Instagram, TikTok) to privacy-friendly frontends (Piped, Nitter, Redlib, SearXNG).
- Minimal-permission architecture published on Chrome Web Store, Firefox Add-ons (incl. Android), and Microsoft Edge.
- Automated instance health checking and sync workflows maintained via GitHub Actions (pritr/instances).

JanSahay — Rural Welfare Eligibility Platform jansahay.pages.dev

- Built an offline-first welfare entitlement assistant designed for rural citizens in Bihar with Devanagari TTS and 48px touch targets.
- Engineered a deterministic rule engine decoupled from LLMs to ensure zero hallucination on benefit eligibility decisions.
- Monorepo deployed on Cloudflare Workers and React PWA with local sahayak verification workflows.

chaind-cli — Sovereign AI Agent Daemongithub.com/fossism/chaind-cli

Lead Developer (Go)2024 - Present

- Headless Go daemon bridging messaging apps (WhatsApp, Matrix, Telegram) directly to local AI agents via permission-gated Unix sockets.
- Designed for secure, zero-cloud interaction with self-hosted LLMs without exposing open network ports.

Experience & Community

Bodhya FOSS Communitybodhya.net — Bihar, India

Tech & Programs Lead — Founding Core TeamJan 2026 - Present

- Lead tech and programs for a FOSS community aligned with FOSS United & Samagata linking tier-2/3 college students with open-source mentors, workshops, and real projects.
- Built production tooling including automated certificate generators, event portals, and Frappe/Listmonk sync integrations (listbrew).

Navprayas — Student-Run Non-Profit Societynavprayas.in — Gaya, Bihar

Team Lead — Website, Database & OperationsMay 2025 - Present

- Directed database operations and registration platforms for educational initiatives serving 15,000+ rural students.
- Handled online registration transitions with Razorpay gateway and anchored annual felicitation ceremonies.

Technical Skills

Languages	Go, Python, TypeScript, JavaScript, C, C++, Bash, SQL
Systems & AI	Model Context Protocol (MCP), Unix Sockets & IPC, AST Program Analysis, Speech DSP & Telephony, CRDTs & Merkle DAGs
Web & Edge	React, Astro, Tailwind CSS, Node.js, Cloudflare Workers, Web Crypto, PWA
Tools & Infra	Linux (Arch), Git/GitHub, Docker, GitHub Actions CI/CD, Nginx, PostgreSQL, Supabase

Education

Government Engineering College (GEC), SheikhpuraSheikhpura, Bihar, India

B.Tech in Computer Science & Engineering — Bihar2024 - 2028

Engineering University (BEU)